
Hybrid Thompson-UCB for Risk-Aware Classification of Brain Tumours under Limited Data

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Abstract: Brain tumour classification under limited MRI data is not just an accuracy problem. It is a risk-aware task because tumour cases misclassified as healthy may lead to false reassurance. This study reframes four-class brain image classification (glioma, meningioma, pituitary, and healthy) as a single-step contextual bandit problem. We propose a framework based on contextual bandits for cost-sensitive classification framework for risk-aware learning. The framework incorporates cost-sensitive learning directly into the reward system, enabling the agent to focus on clinically risky tumour-to-healthy errors and to reduce them by iteratively improving the classification policy. This paper employs hybrid Thompson Sampling with Upper Confidence Bound (TS-UCB), ϵ -greedy, and Boltzmann exploration policies for risk-aware classification with limited brain-tumour data, and evaluates these policies across three data regimes using cross-validation. The hybrid TS-UCB exploration policy achieves the strongest safety-performance balance, obtaining the lowest tumour-to-healthy error rate and the highest healthy-class precision while maintaining competitive validation and test accuracies. The results further demonstrate that reward shaping can tune the safety-accuracy trade-off by prioritising clinically risky confusions while remaining competitive with, and, in some cases, outperforming traditional deep learning techniques under limited data. These results suggest that hybrid TS-UCB is suitable for risk-aware brain tumour classification when labelled data are scarce and false negatives must be controlled.

1. Introduction

Brain Tumour Classification (BTC) is a safety-critical task because different errors have different consequences. In particular, predicting a tumour case as healthy may delay further investigation and treatment. Therefore, this work treats BTC not only as an accuracy problem, but also as a risk-aware decision problem. Conventional classification objectives typically focus on overall predictive performance and may not explicitly distinguish between errors with different levels of consequence. This motivates the use of learning formulations in which different prediction outcomes can be assigned different costs.

Reinforcement Learning-based (RL) methods have been explored in medical imaging by combining reward-based learning with pretrained neural networks. For example, [1] applied Q-learning with transfer-learning backbones for medical image classification, while [2] proposed a deep RL framework for BTC. These studies demonstrate that reward-based decision mechanisms can be integrated with deep visual representations.

Contextual bandits (CB) provide a simpler RL formulation in which the learner observes a context, selects an action, and receives feedback only for the chosen action [3]. In image classification, the context corresponds to image features, while the actions correspond to class labels. This formulation is particularly suitable for cost-sensitive classification because the reward function can assign different values to different

prediction outcomes. In BTC, for example, predicting a tumour image as healthy can be considered more critical than confusing one tumour subtype with another. Prior work has explored asymmetric objectives and reward-shaping methods for bandit-style classification [4–6].

This study proposes a cost-sensitive contextual-bandit-style framework for four-class brain tumour classification: glioma, meningioma, pituitary, and healthy. Each image is represented by a fused deep feature vector, each action corresponds to a class label, and the reward function reflects the relative cost assigned to different prediction outcomes. The reward structure is intentionally asymmetric: correct predictions receive positive rewards, with a larger positive reward assigned to correctly identified *healthy* cases, whereas incorrect predictions receive penalties, with tumour-to-healthy (THR) errors receiving the strongest penalty. This asymmetric design is intended to discourage selection of the Healthy class when the evidence for that decision is insufficient, while still rewarding correct Healthy predictions.

This reward design also motivates the use of safety-oriented evaluation measures in addition to conventional classification metrics. Healthy-class precision (Pr_H) is particularly relevant because, in this four-class setting, a false-positive Healthy prediction corresponds to a tumour case being classified as *healthy*. Thus, improving Pr_H reduces the proportion of tumour cases among the samples predicted as Healthy. Similarly, the THR directly measures the frequency with which tumour cases are incorrectly assigned to the Healthy class. These measures therefore provide complementary information about the reliability of Healthy predictions.

CB and RL methods are also relevant in limited-data settings. RL active learning can reduce labelling requirements [7], and RL models have shown promising performance on small MRI datasets [8–10]. This motivates evaluating contextual-bandit classifiers when training data are scarce. In particular, the most data-limited regime investigated in this work contains only 120 2D MRI images, with 30 images per class, providing a challenging setting for evaluating cost-sensitive learning under severe data scarcity.

The present study uses 2D MRI images because the available dataset is slice-based and the experiments focus on limited-data classification. This also enables direct use of ImageNet-pretrained feature extractors with lower computational and memory requirements than full 3D modelling. Prior work has shown the value of transferring 2D pretrained networks to brain-MRI tasks [11], while a small-scale study reported that a 2D CNN outperformed tested 3D alternatives under limited-data conditions [12]. However, 3D models can exploit inter-slice spatial information, albeit with greater computational and data demands [13]. Therefore, 2D imaging is adopted here as a practical choice consistent with the available data and experimental setting.

Exploration policy is a key component of contextual-bandit learning. Common approaches include ϵ -greedy and Boltzmann exploration, while Thompson Sampling (TS) and Upper Confidence Bound (UCB) provide uncertainty-aware alternatives. Recent work has shown that these principles can be combined. TS-UCB [14], for example, integrates posterior sampling with confidence bounds and improves upon standard TS with little additional computational cost. Such hybrid exploration strategies are appealing in limited-data settings where uncertainty estimation is important.

In this work, we compare two reward-estimation mechanisms combined with three exploration-policy families: ϵ -greedy, Boltzmann exploration, and hybrid TS-UCB. We use linear and neural variants of the estimators, as described in Section 2.2. The models are evaluated across three data regimes using classification accuracy, THR, healthy-class precision (Pr_H), and class-wise recall. The proposed models are also compared with conventional machine-learning and deep-learning classifiers to assess whether the contextual-bandit formulation provides a useful alternative decision framework under limited-data conditions.

Throughout this paper, cost-sensitive refers to the design of the reward function, whereas risk-aware refers to the evaluation of THR errors and the reliability of Healthy predictions in the experimental setting. These terms do not imply clinical validation or formal clinical safety guarantees.

2. Methodology

The proposed framework consists of four stages: preprocessing, deep feature extraction, contextual-bandit-style classification, and Risk-aware evaluation, see Figure 1. The aim is to classify MRI images into four classes while reducing tumour-to-healthy misclassification.

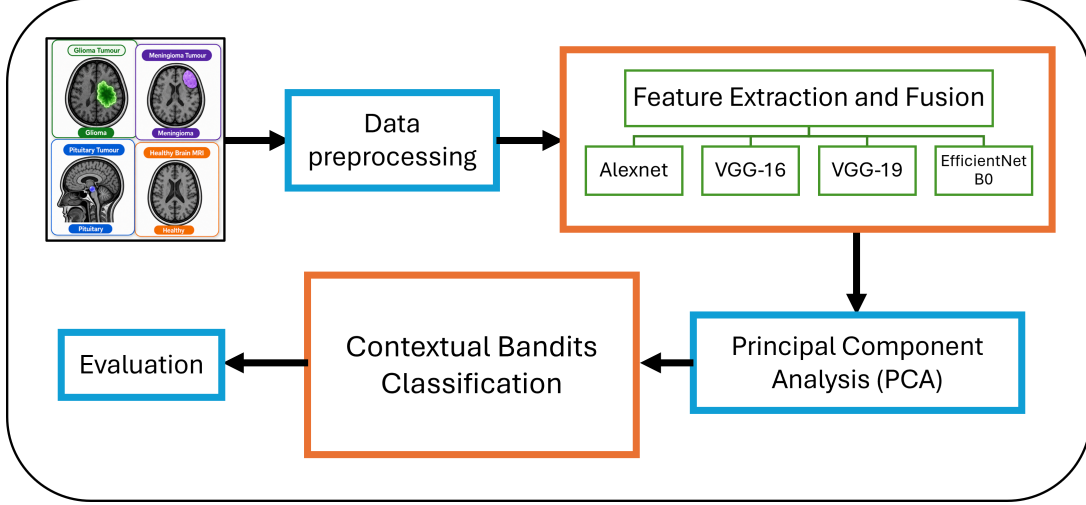


Figure 1. Full pipeline of risk-aware classification with contextual bandits.

2.1. Preprocessing and feature extraction

For each image x_i , four views are generated: the original image, horizontal flip, and rotations by $+15^\circ$ and -15° . Each view is resized to 224×224 , and normalised using ImageNet statistics. Skull stripping is not applied, so the classifier is evaluated on the full MRI view.

Four pretrained convolutional neural network (CNN) backbones are used for feature extraction: AlexNet, VGG16, VGG19, and EfficientNet-B0, implemented in PyTorch using the `torchvision.models` module with ImageNet-pretrained weights. VGG refers to the Visual Geometry Group network family; VGG16 and VGG19 denote architectures containing 16 and 19 weight layers, respectively [15]. The full feature extraction and fusion process can be summarised as

$$\begin{aligned}
 x_i^{(m)} &= \tau_m(x_i), \quad m = 1, \dots, 4, \quad i = 1, \dots, N \\
 z_i^{(m,b)} &= \phi^{(b)}(x_i^{(m)}), \quad b = 1, \dots, 4, \\
 z_i^{(m)} &= z_i^{(m,1)} \oplus z_i^{(m,2)} \oplus z_i^{(m,3)} \oplus z_i^{(m,4)}, \\
 f_i &= \frac{1}{4} \sum_{m=1}^4 z_i^{(m)}, \quad F = [f_1, f_2, \dots, f_N]^\top.
 \end{aligned} \tag{2.1}$$

where $\tau_m(\cdot)$ denotes the image transformation, $\phi^{(b)}(\cdot)$ denotes the b -th CNN backbone, $\oplus$ denotes feature concatenation, f_i is the final fused feature vector for image i , and F is the complete fused feature matrix.

PCA is fitted only on the training split and then applied unchanged to the validation and test splits. The number of retained principal components, denoted by K , is selected to preserve 99.5% of the variance. Therefore, only the first K principal components are used. This reduces the original fused feature vector from D dimensions to K dimensions, where K is determined from the training data for each experimental split.

$$\tilde{\mathbf{f}}_i = \text{PCA}_K(\mathbf{f}_i), \tag{2.2}$$

We also conducted an ablation study which compared shared preprocessing with backbone-specific preprocessing:

$$x_i^{(m,b)} = \pi^{(b)}(A_m(x_i)), \quad z_i^{(m,b)} = \phi^{(b)}(x_i^{(m,b)}). \tag{2.3}$$

Shared preprocessing was selected for the final pipeline because it produced more stable feature alignment and reduced variance across backbones, for more details see (Github).

2.2. Contextual-bandits classification

The classification task is formulated as a contextual-bandit-style decision problem, where the action space is $\mathcal{A} = \{\text{glioma}, \text{meningioma}, \text{pituitary}, \text{healthy}\}$. At each step t , the agent observes a PCA-reduced fused

feature vector $s_t = \tilde{f}_t$ and selects an action $a_t \in \mathcal{A}$, which represents the predicted class.

In this setting, the policy acts as a classifier. It estimates the expected reward for each class and selects the class according to an exploration strategy. This allows the final decision to depend not only on the class's likelihood but also on the cost-sensitive reward structure.

2.2.1 Reward estimation.

Two types of reward estimators are used. The neural estimator is an MLP reward model that outputs one reward score for each arm/class and is trained using one-step MSE loss on the selected arm. The linear estimator uses a separate linear model for each arm. In both cases, scores are computed for all arms during action selection, but only the selected arm is updated after the observed reward is received.

2.2.2 Exploration policies

The reward estimators are combined with three exploration-policy families: ϵ -greedy, Boltzmann exploration, and hybrid Thompson Sampling-Upper Confidence Bound (TS-UCB). Each model name identifies both the reward estimator and the exploration policy. The prefix N- denotes the neural MLP reward estimator, whereas the prefix Lin- denotes the linear reward estimator. Thus, N-TS-UCB and Lin-TS-UCB use the neural and linear reward estimators, respectively, with hybrid TS-UCB exploration. Similarly, N-Boltz and Lin-Boltz use Boltzmann exploration, and N- ϵ -Greedy and Lin- ϵ -Greedy use ϵ -greedy exploration.

For Boltzmann exploration, actions are sampled according to

$$\pi(a \mid s_t) = \frac{\exp(\hat{r}(s_t, a)/\tau)}{\sum_{a' \in \mathcal{A}} \exp(\hat{r}(s_t, a')/\tau)}, \quad (2.4)$$

where $\hat{r}(s_t, a)$ is the estimated reward for action a in context s_t , and τ is the temperature parameter. This is the standard Boltzmann (or softmax) action-selection rule used in the reinforcement learning and bandit literature [16]. A lower value of τ concentrates selection on actions with higher estimated reward, whereas a larger value increases exploration.

The hybrid TS-UCB policy combines Thompson Sampling, originally proposed by Thompson [17], with the Upper Confidence Bound (UCB) principle introduced for multi-armed bandits by Auer et al. [18]. Thompson Sampling selects actions according to samples drawn from a posterior reward distribution, whereas UCB augments estimated rewards with an uncertainty-dependent confidence bonus to encourage exploration. Several hybrid variants of TS-UCB have been proposed in the contextual-bandit literature, combining posterior sampling with confidence-bound exploration mechanisms [19].

2.3. Cost-sensitive reward system

The contextual-bandit (CB) setting is simulated using labelled data by revealing only the reward of the selected action during training. The reward is manually shaped using an observed confusion matrix, which assigns the largest penalty to THR errors, as these errors correspond to tumour images predicted as healthy.

A compact reward function is defined as:

$$r_t = \begin{cases} r_H, & a_t = y_t = H, \\ r_C, & a_t = y_t, y_t \in T, \\ r_W - \lambda_{y_t}, & a_t = H, y_t \in T, \\ r_W, & a_t \neq y_t, a_t \neq H, \end{cases} \quad (2.5)$$

where H denotes the healthy class, $T = \{G, M, Pit\}$ denotes the tumour classes, and $(r_H, r_C, r_W) = (2.0, 0.9, -0.5)$ are denoting reward of true healthy prediction, reward of any other correct prediction, and the penalty of generic wrong prediction, respectively. The tumour-specific false-negative penalty is $\lambda_G = 1.90$ and $\lambda_M = \lambda_P = 1.25$. Lastly, a_t and y_t are the action taken at step t , and the true label at step t , respectively.

This reward structure encourages the policy to avoid predicting healthy unless the expected reward supports that decision. λ_{y_t} is intended to encode *clinical harm*, not class-specific prediction difficulty. The

ordering $\lambda_G > \lambda_M = \lambda_P$ reflects the greater potential harm of missing an aggressive glioma: most meningiomas are WHO (World Health Organization) grade 1, and asymptomatic cases may be managed through observation [20], while pituitary tumours are likewise predominantly benign and clinically indolent [21], with only 8% enlarging over approximately three years and managed by follow-up MRI scans [22] rather than resection [23]. In contrast, glioblastoma can double in volume approximately monthly [24].

As shown in Section 2.5, meningioma has the lowest recall in Table 2, raising the question of why λ_M is not larger than λ_G . Although additional experiments indicate that increasing λ_M may indirectly improve overall accuracy and meningioma recall while reducing the THR, it does not directly address the source of the recall deficit. To justify that, in Eq. (2.5), λ_{y_t} applies only when $a_t = H$; it therefore enters the training signal only following a healthy prediction and updates the healthy arm. Consequently, increasing λ_M can make the policy more conservative in classifying meningioma as healthy, but it does not directly reshape the tumour decision boundaries, because most meningioma errors are confusions with other tumour classes rather than healthy predictions, as the experiment shows. Additionally, the same meningioma recall deficit appears even in the CNN baselines, which use no reward matrix, suggesting that it primarily reflects classifier difficulty rather than the reward design.

Table 1. Dataset B and its sub-datasets, see Sect. 2.4.

Name	Source	Size	Classes	Notes
B	Figshare [25] + SAR-TAJ [26] + Br35H [27]	7,023	4	Cleaned, relabelled glioma
BC	Subset of B	360 (90×4)	4	Balanced
B2	Subset of B	120 (30×4)	4	Small

2.4. Datasets

The dataset B [25] is used in different settings. It contains MRI brain scans of four classes: glioma, meningioma, no tumour, and pituitary. The dataset is used in three different settings (see Table 1). Each run splits into 4-fold cross-validation (60% train, 20% validation, 20% test).

2.5. Experiments

The experiments evaluate whether hybrid Thompson Sampling-Upper Confidence Bound (TS-UCB) provides a risk-aware policy under limited data. Focusing on Dataset B2, the main evaluation criteria are tumour-to-healthy ($T-H$) rate, healthy-class precision (Pr_H), tumour-class recall, and validation and test accuracies. Table 2 reports the compact classification results on Dataset B2, for more details on the full experiment (Github).

On dataset B, the lowest THR and highest Pr_H are achieved by AlexNet, followed by VGG16, VGG19, EfficientNet, and then N-TS-UCB. Among the contextual-bandit models, N-TS-UCB ranked best on the safety-oriented metrics, followed by LiN-TS-UCB and Lin- ϵ -Greedy.

For dataset BC, VGG16 achieves the best safety-oriented performance, with the lowest THR and the highest Pr_H . The N-TS-UCB is the best contextual-bandit model, ranking second overall with a THR of 0.0207 and Pr_H of 0.940. VGG19 and N-Boltz ($\tau = 2.0$) showed similarly strong safety results, followed by VGG19 and N-Boltz ($\tau = 2.0$), while Lin- ϵ -Greedy and EfficientNet showed the weakest results on these safety-related measures.

Table 2. Summary of Model Classification Performance on Dataset B2 in ascending order from lowest THR and High Healthy class Precision.

Model	Valid acc	Test acc	THR	(H) Pr	(G) R	(M) R	(Pit) R
N-TS-UCB	0.914	0.836	0.0413	0.884	0.879	0.743	0.948
VGG19	0.785	0.743	0.0422	0.877	0.827	0.537	0.850
N-Boltz ($\tau = 1.5$)	0.907	0.825	0.0457	0.866	0.900	0.657	0.969
N- ϵ -Greedy	0.901	0.805	0.0543	0.851	0.900	0.551	0.962
VGG16	0.801	0.794	0.0611	0.851	0.897	0.597	0.860
N-Boltz ($\tau = 2.0$)	0.896	0.836	0.0630	0.834	0.914	0.680	0.910
AlexNet	0.774	0.745	0.0722	0.806	0.823	0.527	0.877
Lin-TS-UCB	0.875	0.800	0.0935	0.769	0.866	0.554	0.916
Lin-Boltz ($\tau = 1.5$)	0.868	0.810	0.1022	0.765	0.868	0.568	0.916
Lin-Boltz ($\tau = 2.0$)	0.868	0.810	0.1022	0.765	0.868	0.568	0.916
Lin- ϵ -Greedy	0.884	0.760	0.1109	0.731	0.844	0.509	0.857
EfficientNet	0.626	0.620	0.2244	0.595	0.807	0.163	0.610

Note. H = Healthy, G = Glioma, M = Meningioma, Pit. = Pituitary. Only the main safety-related metrics are shown; other metrics, along with their standard deviations, are on the (Github) link.

3. Results

3.1. Safety-performance trade-off and class-wise analysis

The results show that the best model is not the one with the highest overall accuracy as shown in section 2.5. As summarised in Table 3, the strongest case is when a model combines high accuracy, high healthy-class precision, and a low THR. This pattern appears clearly in dataset B, where AlexNet achieved the best safety profile with a THR of 0.0058 and Pr_H of 0.985, while still maintaining high test accuracy. A similar pattern is seen in dataset BC, where VGG16 achieved the lowest THR of 0.0189 and the highest Pr_H of 0.945.

However, dataset B2 shows why accuracy must be interpreted together with the safety metrics. N-TS-UCB and N-Boltz ($\tau = 2$) achieved the same test accuracy of 0.836, but N-TS-UCB had a lower THR of 0.0413 and a higher Pr_H of 0.884, compared with 0.0630 and 0.834 for N-Boltz ($\tau = 2$). According to the trade-off interpretation in Table 3, N-TS-UCB therefore represents the safer choice, even when the two models have equal test accuracy.

The weaker cases also support this interpretation. For example, EfficientNet on dataset B2 produced the lowest test accuracy of 0.620, the highest THR of 0.2244, and the lowest Pr_H of 0.595. Similarly, Lin- ϵ -Greedy showed weaker safety-oriented behaviour on dataset BC, with the highest THR of 0.0526 and the lowest Pr_H of 0.859. These examples show that the safest model is the one that maintains accuracy while also reducing THRs and improving the reliability of healthy predictions. See 2.5 for the model ranking at each dataset according to the lower THR.

Table 3. Interpretation of the Trade-off Between Accuracy, Healthy-Class Precision, and THR.

Observed Pattern	Meaning	Interpretation
High accuracy (Pr_H), low THR	Best overall case	Accurate and clinically safer
High accuracy (THR) and low Pr_H	Unsafe despite good accuracy and Healthy predictions are less reliable	Risk of false reassurance and Healthy outputs need caution
High Pr_H and low THR	Strong safety-aware behaviour	Healthy predictions are trustworthy

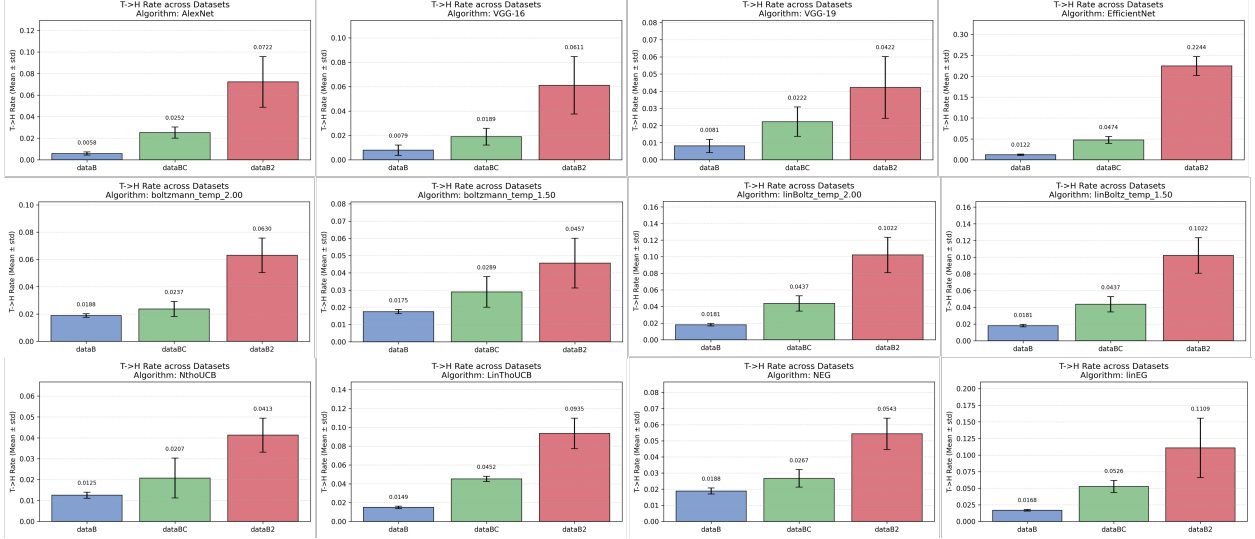


Figure 2. THR across dataset B, dataset BC, and dataset B2 for each model. Bars show the mean THR, and error bars show the standard deviation. The figure shows that all models become increasingly vulnerable to T-H errors as dataset size decreases, but the rate of degradation differs across models.

3.2. Stability of THR across dataset shifts

Figure 2 and Table 4 show how the THR changes when the dataset size decreases from dataset B to dataset BC and then to dataset B2.

To clarify the comparison in Table 4, the reported differences are calculated between the tumour-to-healthy misclassification rates, not between the datasets themselves. Let $R_m(B)$, $R_m(BC)$, and $R_m(B2)$ denote the mean THR of model m on Dataset B, Dataset BC, and Dataset B2, respectively. The first jump $J_1(m)$ measures the increase in THR from Dataset B to Dataset BC, and the second jump $J_2(m)$ measures the increase in THR from Dataset BC to Dataset B2:

$$J_1(m) = R_m(BC) - R_m(B), \quad J_2(m) = R_m(B2) - R_m(BC).$$

The steadiness gap of the model $G(m)$ measures how different these two increases are:

$$G(m) = |J_2(m) - J_1(m)|.$$

A smaller steadiness gap means that the model’s THR increases more smoothly as the dataset size decreases.

Overall, the THR increases for all models as the dataset became smaller, but the increase was not uniform. For instance, Boltzmann exploration at $T = 1.50$ shows the smoothest growth, with the smallest steadiness gap of 0.0054. Its THR increases gradually from 0.0175 on dataset B to 0.0289 on dataset BC and 0.0457 on dataset B2. Another observation tells that VGG19 also shows very stable behaviour, with a small steadiness gap of 0.0059 and a gradual increase from 0.0081 to 0.0222 and then 0.0422. This indicates that some models preserve relatively consistent safety behaviour even when the number of samples is reduced. In addition, N-TS-UCB shows a moderate but still favourable pattern. Its THR increased from 0.0125 on dataset B to 0.0207 on dataset BC and 0.0413 on dataset B2, with a steadiness gap of 0.0124. Although it is not the smoothest model, it achieves the lowest THR on dataset B2, which makes it the strongest safety-aware model in the small-data setting. This suggests that stability and final safety performance should be interpreted together: a model can be slightly less steady across shifts but still become the safest model on the smallest dataset, like N-TS-UCB.

In contrast, some models degraded sharply when moving to dataset B2. For example, EfficientNet shows the most dramatic increase, with the THR rising from 0.0122 on dataset B to 0.0474 on dataset BC and 0.2244 on dataset B2. Similarly, linBoltz at both temperatures shows a clear second jump, increasing from

0.0437 on dataset BC to 0.1022 on dataset B2.

These results indicate that robustness under reduced data is model-dependent and should not be judged from single-dataset accuracy alone.

Table 4. Steadiness of T→H rate growth across datasets

Rank	Algorithm	dataB	dataBC	dataB2	First Jump	Second Jump	Steadiness Gap	Interpretation
1	N-Boltz ($\tau = 1.5$)	0.0175	0.0289	0.0457	0.0114	0.0168	0.0054	Smoothest growth
2	VGG19	0.0081	0.0222	0.0422	0.0141	0.0200	0.0059	Very steady
3	N-TS-UCB	0.0125	0.0207	0.0413	0.0082	0.0206	0.0124	Moderate growth
4	LiN-TS-UCB	0.0149	0.0452	0.0935	0.0303	0.0483	0.0180	Steady but higher increase
5	N- ϵ -Greedy	0.0188	0.0267	0.0543	0.0078	0.0277	0.0199	Second jump larger
6	Lin- ϵ -Greedy	0.0168	0.0526	0.1109	0.0358	0.0583	0.0224	Smooth but larger growth
7	AlexNet	0.0058	0.0252	0.0722	0.0194	0.0470	0.0277	Moderate-to-large second jump
8	VGG16	0.0079	0.0189	0.0611	0.0110	0.0422	0.0312	Large second jump
9	Lin-Boltz ($\tau = 1.5$)	0.0181	0.0437	0.1022	0.0256	0.0585	0.0328	Clear second jump
10	Lin-Boltz ($\tau = 2.0$)	0.0181	0.0437	0.1022	0.0256	0.0585	0.0328	Clear second jump
11	N-Boltz ($\tau = 2.0$)	0.0188	0.0237	0.0630	0.0049	0.0393	0.0345	Large second jump
12	EfficientNet	0.0122	0.0474	0.2244	0.0352	0.1770	0.1419	Most dramatic increase

4. Conclusion

This study investigates a cost-sensitive contextual bandit framework for risk-aware brain tumour classification under limited data conditions. By assigning asymmetric rewards to different prediction outcomes, the framework explicitly prioritises the reduction of THR while maintaining competitive classification performance. Across the three data regimes, the results show that risk-aware performance cannot be assessed from accuracy alone, since models with similar accuracies can differ in the Tumour-to-Healthy error rate (THR) and Healthy-class precision (Pr_H).

The strongest benefit of the proposed approach appears in Dataset B2. N-TS-UCB achieves a validation accuracy of 0.914 and a test accuracy of 0.836, together with the lowest THR(0.0413) and the highest Pr_H (0.884) among the evaluated models, providing the strongest safety-performance balance in this setting. N-TS-UCB also shows a relatively controlled increase in THR as the dataset size decreases. However, the results do not indicate that the contextual bandit model generally outperforms conventional CNN classifiers: AlexNet and VGG16 achieve the strongest safety-oriented results on Datasets B and BC, respectively. Rather, the findings suggest that hybrid TS-UCB (i.e. N-TS-UCB) is particularly useful when labelled data are limited and asymmetric prediction costs are incorporated into the learning process.

The experiments also highlight several limitations. Performance remains dependent on the quality of the extracted representations and on class-specific difficulty, with meningioma recall remaining lower than that of the other tumour classes in Dataset B2. In addition, the study uses 2D slice-based MRI data and manually defined reward values; therefore, they do not constitute clinical validation or formal clinical safety guarantees. Future work will investigate patient-level volumetric MRI, improved medical feature representations, targeted methods for difficult tumour classes, automated or data-driven reward optimisation, and clinician-in-the-loop decision mechanisms for risk-aware classification [28]. These extensions will help determine whether the observed risk-aware behaviour of the contextual bandit framework generalises to larger and more clinically representative datasets.

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